

Ovulation



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Definition

Ovulation is the physiological process in which a mature egg (ovum) is discharged from one of the ovaries into the fallopian tube, where it can potentially be fertilized by sperm. It is a critical stage in the menstrual cycle of female mammals, including humans. It usually occurs around the 14th day of a 28-day cycle; however, it can vary from person to person. A number of morphological and biochemical processes take place inside the preovulatory follicle during the ovulation process. In the ovarian environment, various genes get activated under the influence of gonadotropins and sex steroids which are controlled by various growth factors resulting in enzymatic and structural changes. All of those actions increase the likelihood that the oocyte will be fertilized and extruded on the ovarian surface to form the corpus luteum (Russell & Robker, 2007).

Ovulation Process

“Ovulation” is the part of reproductive cycle. As such the reproductive cycle is divided into three phases: (1) Menstruation and the follicular phase, (2) ovulation phase, and (3) the luteal phase. These three phases refer to the status of the ovary during the reproductive cycle.

Menstruation and the Follicular Phase

When there is no pregnancy, the menstrual phase (also known as menstruation or the menstrual cycle) is characterized by the shedding of the uterine lining (endometrium) in the form of menstrual bleeding, the first day of which is considered day 1 of the menstrual cycle. It typically lasts around 3–7 days, though this can vary among individuals. When conception does not occur, menstruation is initiated by a drop in the levels of the hormone’s estrogen and progesterone, which triggers the breakdown and shedding of the endometrial tissue. The follicular phase follows menstruation and typically lasts from the end of menstruation until ovulation. During the follicular phase, the anterior pituitary gland releases follicle-stimulating hormone (FSH) in response to signals from the hypothalamus (Richards et al., 1998). FSH stimulates the growth and development of ovarian follicles, each containing an immature egg (oocyte). Only one dominant follicle will usually mature during this phase. As

the dominant follicle grows, it produces increasing amounts of estrogen. This rise in estrogen levels leads to the thickening of the uterine lining in preparation for potential implantation of a fertilized egg (Marieb, 2013). Even while numerous follicles start the maturation process, only the follicle with the most granulosa cells, FSH receptors, and estrogen synthesis becomes the dominant follicle, and the nondominant follicles develop atresia.

Ovulation Phase

The follicular phase ends with ovulation, which is triggered by a surge in luteinizing hormone (LH) and a further increase in estrogen levels. Ovulation is the release of a mature egg from the dominant follicle, making the woman fertile. Ovulation normally happens in the midst of a regular 28-day menstrual cycle, or around the 14th day. However, the timing can differ between people and cycles. The quick process of ovulation typically lasts for 24 h. Ovulation is marked by an increase in LH (LH surge), which is secreted by the pituitary gland in response to rising estrogen levels. The LH surge begins 34–36 h prior to ovulation, and peak LH secretion occurs 10–12 h prior to ovulation. This increase in LH leads the mature ovarian follicle (which contains the egg) to rupture and the egg to be released from the ovary (Watson & Stacy, 2010). When LH levels rise, the granulosa and theca cells undergo several changes and begin producing progesterone. After the LH surge and the release of the first polar body, meiosis of the primary follicle resumes; the oocyte then pauses in metaphase of the second meiotic division until fertilization occurs. Some women experience pelvic pain or discomfort during ovulation around the midpoint of their menstrual cycle which is known as *Mittelschmerz* (a German term that translates to “middle pain” or “pain in the middle”). *Mittelschmerz* is mostly considered a normal and benign in occurrence, is generally a self-limiting condition, and does not require treatment in most cases (Richards et al., 1998).

Luteal Phase

It occurs after ovulation and before menstruation, and it is distinguished through distinct hormonal changes and anatomical changes in the female reproductive system. This phase usually occurs after ovulation and can last for approximately 10–16 days. The length of this phase varies from person to person and cycle to cycle. Furthermore, during this phase, there are notable hormonal shifts. After ovulation, the ruptured follicle in the ovary transforms into a structure called the corpus luteum. The corpus luteum begins to produce the hormone progesterone, which plays a central role in this phase. Progesterone is responsible for maintaining the uterine lining (endometrium) and preparing it for the potential implantation of a fertilized egg. Follicular development has resulted in an increase in the number of LH receptors on granulosa and theca cells. These LH receptors are activated by the midcycle LH surge, which converts their enzymatic machinery to make and secrete progesterone; this is known as luteinization. During the luteal phase, as progesterone inhibits pituitary secretion of FSH and LH, both the hormones are inhibited. The corpus luteum also secretes estradiol in a pattern similar to progesterone secretion. Progesterone production begins about 24 h before ovulation and increases significantly after that. Progesterone production peaks 3–4 days after ovulation. The corpus luteum has a lifespan of 9–11 days after ovulation; if conception does not occur, the corpus luteum undergoes involution (a progressive decrease in size), and progesterone production reduces drastically. Progesterone withdrawal releases FSH from negative feedback, which causes levels to rise before menstruation and the start of a new cycle. Under the influence of progesterone, the endometrium gains thickness, becomes more vascularized, and enriched with nutrients, creating a favorable environment for the possible implantation of a fertilized egg. If the oocyte is fertilized and implantation takes place, the zygote starts secreting human chorionic gonadotropin (hCG), which sustains the corpus luteum for another 6–7 weeks. Early pregnancy

requires adequate progesterone production by the corpus luteum. Placental steroidogenesis is well established by 9–10 weeks of pregnancy, and the placenta takes over progesterone synthesis (Richards et al., 1998).

Hormones Involved in the Process of Ovulation

A complicated interaction of hormones produced by the hypothalamus, pituitary gland, and ovaries regulates ovulation. These hormones collaborate to initiate and control the ovulation process. The following hormones are important in regulating ovulation:

Gonadotropin-Releasing Hormone (GnRH), also known as Luteinizing Hormone-Releasing Hormone (LHRH), is a tropic peptide hormone synthesized and secreted by the hypothalamus from GnRH neurons. Vertebrates have approximately 20 major structures of GnRH and its receptor, which indicates that the GnRH system evolved early in the evolutionary process (Millar, 2005). It is a neurohormone made up of 10 amino acids that is produced in the hypothalamic arcuate nuclei. The preoptic region of the hypothalamus, which houses the majority of the GnRH-secreting neurons, is a crucial region for GnRH synthesis (Okubo & Nagahama, 2008). It is a releasing hormone that stimulates the release of FSH and LH from the anterior pituitary gland. Two forms of GnRH secretion have been identified, the pulsatile and surge forms. In the pulsatile mode, GnRH is released episodically in discrete pulses into the portal circulation, with undetectable GnRH concentrations in between pulses (Maeda et al., 2010). In females, the surge mode of GnRH secretion occurs during the pre-ovulatory phase, during which GnRH appears to be persistent in the portal circulation (Maeda et al., 2010; Moenter et al., 2003). Under ordinary circumstances, GnRH is released in pulses at 90- to 120-min intervals. The releasing hormone must be provided in pulses to patients with GnRH insufficiency in order to enhance

serum gonadotropin concentrations. Contrarily, ongoing GnRH therapy reduces gonadotropin output, which has therapeutic advantages for some patients, including children experiencing premature puberty and men with prostate cancer (Moenter et al., 2003). The secretion of GnRH, FSH, and LH is regulated by complex feedback mechanisms involving the levels of sex hormones, primarily estrogen and progesterone. These hormones provide feedback to the hypothalamus and pituitary gland, influencing the timing and intensity of GnRH pulses and subsequent hormonal releases. GnRH is a key regulator of the female menstrual cycle and male sperm production. It is essential for ovulation timing, uterine lining preparation for prospective implantation, and the maintenance of normal reproductive function. GnRH and its synthetic analogs are used in medical treatments. For example, GnRH agonists and antagonists are employed in assisted reproductive technologies, such as in vitro fertilization (IVF) and the management of conditions like endometriosis and uterine fibroids. GnRH analogs are also used to suppress gonadal function in conditions like precocious puberty and certain hormone-dependent cancers.

Gonadotropins are a group of hormones that play a crucial role in regulating the growth, development, and function of the gonads. Gonadotropins are heterodimeric glycoproteins with two sub units alpha/beta. The two primary gonadotropins are follicle-stimulating hormone (FSH) and luteinizing hormone (LH). Another type of gonadotropin produced in women during pregnancy by the placenta is hCG (Kara et al., 2019).

Follicle-stimulating hormone (FSH): FSH is a glycoprotein heterodimer that consists of alpha and beta subunits. The beta subunit is specific to FSH, whereas the alpha subunit is shared by hCG and LH (Barbieri, 2014). FSH promotes the maturation of primordial germ cells in both males and females. In females, before ovulation, FSH promotes the development and maturation of immature oocytes into mature (Graafian) secondary follicles. The granulosa

cells that encircle growing ovarian follicles have G-protein-coupled receptors called FSH receptors (Richards & Pangas, 2010). By growing follicles, FSH also promotes the production of estradiol. When the follicle grows to a diameter of 8–10 mm, it begins to secrete large levels of estradiol. Normally, only one follicle in a cohort becomes dominant and survives to grow to 18–30 mm in size and ovulate, while the remaining follicles in the cohort experience atresia. Rise in serum estradiol levels reduces FSH synthesis by blocking GnRH production in the hypothalamus (Dickerson et al., 2008). The actions of FSH in promoting follicle growth and estrogen production are crucial for a woman's fertility. Without FSH, the development of ovarian follicles and the menstrual cycle would not proceed normally.

In males, FSH stimulates the Sertoli cells in the testes, which support the development of sperm cells (spermatogenesis). FSH is essential for the initiation and maintenance of sperm production. FSH also promotes the development of the tubules of the testes. While LH primarily regulates the production of testosterone in males, FSH also has a minor role in supporting testicular function and the production of inhibin, which regulates FSH levels through a feedback loop (Boulpaep & Boron, 2005).

Luteinizing hormone (LH): LH is a glycoprotein heterodimer, the structure being similar to that of the other glycoprotein hormones, follicle-stimulating hormone (FSH), thyroid-stimulating hormone (TSH), and hCG. The beta subunit is specific to LH, whereas the alpha subunit is shared by other glycoproteins (Barbieri, 2014). LH has several functions that differ between men and women. LH aids in the maturation of primordial germ cells in both sexes. LH binds specific transmembrane receptors localized primarily in the testes and ovary. In females, in the middle of the menstrual cycle, around 24–36 h before ovulation LH surge occurs, that acts as a signal to trigger the final stages of ovum maturation and release (ovulation). Monitoring the LH surge can be

helpful for women trying to conceive. After ovulation, LH helps transform the ruptured ovarian follicle into a temporary endocrine structure called the corpus luteum. The corpus luteum produces progesterone, which is important for maintaining the uterine lining during the second half of the menstrual cycle in preparation for a possible pregnancy (Barbieri, 2014).

In males, LH stimulates the Interstitial Cells of Leydig in the testes to produce testosterone by activating the LH/choriogonadotropin receptor (LHCGR), a G-protein-coupled receptor located on Leydig cell plasma membranes, hence also been called interstitial cell-stimulating hormone (ICSH) (Louvet et al., 1975). Testosterone is the primary male sex hormone and is essential for sperm production (spermatogenesis) and the development of secondary sexual characteristics, such as facial hair, deepening of the voice, and muscle development.

Factors affecting Ovulation: Ovulation is a complex and tightly regulated process in females, influenced by a variety of factors, including hormonal, environmental, genetic, and lifestyle factors (Wilcox et al., 1999). Here are some of the key factors that can affect ovulation in females:

Hormonal Factors:

- (a) **Follicle-Stimulating Hormone (FSH) and Luteinizing Hormone (LH):** These two hormones are essential for ovulation. FSH stimulates the growth and development of follicles in the ovaries, while LH triggers the release of the mature egg during ovulation. Imbalances in these hormones can disrupt ovulation.
- (b) **Estrogen and Progesterone:** These hormones play critical roles in regulating the menstrual cycle and ovulation. Fluctuations in estrogen and progesterone levels can impact the timing and regularity of ovulation (Dozortsev & Diamond, 2020).
- (c) **Thyroid Hormones:** Thyroid disorders, such as hypothyroidism or hyperthyroidism, can affect ovulation by disrupting the balance of hormones in the body.

Age: Ovulation can be influenced by age. Ovarian function declines with age, and women may experience irregular ovulation or anovulation (lack of ovulation) as they approach perimenopause and menopause.

Polycystic Ovary Syndrome (PCOS): PCOS is a common hormonal disorder that can disrupt ovulation. Women with PCOS often have elevated levels of androgens (male hormones) and insulin resistance, which can interfere with normal ovulatory processes.

Stress: High levels of stress, both physical and emotional, can disrupt the hypothalamic-pituitary-ovarian axis, leading to irregular ovulation or anovulation.

Body Weight: Both obesity and extreme thinness (underweight) can disrupt ovulation. Changes in body fat levels can affect hormone production and signal the body to stop ovulating.

Exercise: Intense physical activity, particularly in the case of athletes, can lead to irregular ovulation or amenorrhea (absence of menstruation). Excessive exercise can lower body fat levels and disrupt hormone production.

Diet and Nutrition: Poor nutrition, extreme dieting, and eating disorders can impact ovulation by altering hormone levels. Adequate intake of essential nutrients is crucial for reproductive health.

Medications and Drugs: Certain medications, including some birth control methods and medications for chronic conditions, can affect ovulation. Recreational drugs and excessive alcohol consumption can also disrupt hormonal balance.

Medical Conditions: Underlying medical conditions such as diabetes, polyps, fibroids, and ovarian cysts can affect ovulation. Pelvic inflammatory disease (PID) or other infections can also interfere with normal ovarian function.

Environmental Factors: Exposure to environmental toxins, chemicals, and pollutants can disrupt hormonal balance and potentially impact ovulation.

Hormonal Birth Control: The use of hormonal contraceptives, such as birth control pills,

patches, or hormonal IUDs, can suppress ovulation as part of their contraceptive mechanism.

Breastfeeding: Exclusive breastfeeding can suppress ovulation in some women for several months after childbirth, acting as a natural form of contraception.

Genetic factors: While genetics play a role in determining the timing and regulation of ovulation, it is essential to understand that ovulation is a multifactorial event, and no single gene is solely responsible for it.

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